

Controllable Text Generation via Diffusion Models on the E2E NLG Benchmark

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Abstract. We study controllable text generation with continuous-space diffusion language models and compare them against autoregressive baselines on the E2E NLG benchmark. Our implementation of Diffusion-LM combines four design choices that we found necessary for stable training at small scale: prefix conditioning with embedding clamping at every denoising step, target-masked losses that exclude prefix and padding positions, initialization of the token embedding table from a pretrained BERT tokenizer, and an exponential moving average (EMA) of the denoiser weights. On top of this base model we add self-conditioning, which feeds the previous x_0 prediction back into the denoiser, and an attribute classifier trained jointly with the diffusion loss that provides classifier-guided sampling over the `food` slot. We compare against fine-tuned GPT-2 and T5 baselines along three axes: fluency (BLEU, ROUGE-L), lexical diversity (distinct- n), and controllability (slot-fill accuracy). Ablations isolate the contributions of noise schedule (sqrt vs. cosine), DDIM sampling step count, and prefix length. We discuss the trade-off between hard prefix clamping and soft classifier guidance, the practical limits of small-scale continuous diffusion for natural language generation, and the extent to which the heavily templated structure of E2E NLG itself favors autoregressive beam search over iterative refinement.

Keywords: Diffusion models · Controllable generation · Non-autoregressive generation · Classifier guidance · E2E NLG · Natural language generation.

1 Introduction

Autoregressive (AR) language models dominate modern text generation, but their left-to-right decoding ties attribute control to ad-hoc prompt engineering and tends to under-explore the output distribution, hurting lexical diversity. Diffusion models, which have transformed image synthesis [8,14], offer a non-autoregressive alternative that operates in a continuous embedding space and supports plug-and-play guidance [10]. The diffusion formulation iteratively refines an entire sequence in parallel, which makes attribute conditioning both more local (a single position can be clamped) and more global (gradients from an attribute classifier reach every position simultaneously).

This project implements a Diffusion-LM variant with prefix conditioning on the E2E NLG benchmark [15] and asks: *can a small, trainable-from-scratch diffusion language model match autoregressive baselines on a slot-driven generation task while improving the diversity-controllability trade-off?*

Contributions.

1. A from-scratch Diffusion-LM implementation with three interchangeable noise schedules (sqrt, cosine, linear), prefix-clamped denoising diffusion implicit model (DDIM) sampling [19], exponential moving average (EMA) weights, and BERT-initialized token embeddings.
2. Two controllability mechanisms in one model: *hard* prefix clamping that fixes the meaning representation throughout denoising, and a *soft* jointly trained attribute classifier used for gradient-based guidance [4] on a chosen slot.
3. Self-conditioning [1] as an architectural addition that lets the denoiser refine its own previous x_0 prediction across consecutive DDIM steps.
4. A side-by-side comparison with fine-tuned GPT-2 and T5 baselines on E2E NLG using fluency (BLEU, ROUGE-L), lexical diversity (distinct- n), and controllability (slot-fill accuracy).
5. Ablations isolating the effect of noise schedule, DDIM step count, and prefix length.

2 Related Work

Diffusion language models. Diffusion-LM [10] introduced continuous-space text diffusion: tokens are mapped to learned embeddings, noise is added in embedding space according to the sqrt schedule, and a transformer denoiser predicts the clean embeddings (x_0 -prediction). At inference, a rounding step projects denoised vectors back to the vocabulary. Subsequent work extended the framework to sequence-to-sequence tasks (DiffuSeq [6]) and to semi-autoregressive decoding with simplex representations (SSD-LM [7]). Our implementation follows the original Diffusion-LM closely, with the additions described in Section 3.

Self-conditioning. Chen et al. [1] proposed self-conditioning for discrete diffusion: during training, with probability $\frac{1}{2}$, a stop-gradient x_0 estimate from a first forward pass is concatenated to the denoiser input on a second pass. The cost is one extra forward pass during training and a single learned projection; the benefit is that consecutive denoising steps can share information across iterations rather than starting from scratch each time.

Classifier guidance. Dhariwal and Nichol [4] showed that an auxiliary classifier trained on noisy inputs can steer diffusion sampling toward a target class via $x_t \leftarrow x_t + s \nabla_{x_t} \log p_\phi(y | x_t)$. We train such a classifier jointly with the denoiser, using a pooled target-mask average over x_t , and apply the gradient with prefix positions zeroed out so that the meaning representation is not perturbed.

Controllable AR generation. PPLM [2] and FUDGE [20] impose attribute constraints on autoregressive decoding through external classifiers. The diffusion analogue is structurally simpler: gradients flow back through the denoiser at every step rather than only modifying the next-token logits.

E2E NLG benchmark. E2E NLG [15] pairs restaurant meaning representations (MRs) with natural language descriptions. We use the cleaned version [5], which addresses slot-noise issues in the original release.

3 Methodology

3.1 Diffusion-LM with Prefix Conditioning

Token ids $w_{1:L}$ are embedded as $x_0 = E[w_{1:L}] \in \mathbb{R}^{L \times d}$, with $d=768$. The forward process injects Gaussian noise according to the chosen schedule:

$$q(x_t | x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) I). \quad (1)$$

We adopt the sqrt schedule of [10] as our default: $\bar{\alpha}_t = 1 - \sqrt{t/T + s}$ with $s=10^{-4}$ and $T=2000$, and ablate against cosine [14] and linear [8] schedules. A transformer denoiser $f_\theta(x_t, t)$ with six layers, eight heads, and hidden width 512 is trained to predict x_0 .

Prefix conditioning. The first $P=32$ positions hold the tokenized MR (a flattened “key: value | key: value” string). We keep these positions at x_0 in the forward process (no noise added) and exclude them from the loss. At sampling time we clamp the prefix embeddings to the MR embeddings at every DDIM step, so the denoised target is conditioned on the MR throughout. This is the diffusion analogue of inpainting in vision models, and provides controllability *by construction*: the model is never exposed to a noisy MR and cannot drift away from it.

Target masking. Both losses are evaluated only at target positions ($P < i \leq L$) and only on non-padding tokens. Padding positions are masked out of both the MSE and the rounding cross-entropy, which we found necessary for stable training: without the mask, the model spends capacity learning to “denoise” pad tokens that carry no information.

BERT-initialized embeddings. The token embedding table E is initialized from the `bert-base-uncased` [3] input embeddings rather than randomly. The denoiser only ever sees embedding-space inputs, so warm-starting the embedding space from a model that already has meaningful token geometry gives the rounding objective a head start. The embeddings are then trained jointly with the rest of the network.

3.2 Loss

We combine the embedding-space MSE with an auxiliary rounding cross-entropy on the target positions only:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{t, x_0, \epsilon} [\mathbf{1}_{\text{tgt}} \odot (\|x_0 - f_\theta(x_t, t)\|^2 + \text{CE}(W^\top f_\theta(x_t, t), w))], \quad (2)$$

where W is tied to the input embedding matrix E to share parameters.

3.3 Self-Conditioning

At training time, with probability $\frac{1}{2}$, we first run a stop-gradient forward pass $\tilde{x}_0 = f_\theta(x_t, t)$, clamp its prefix positions back to x_0 , and then take a second pass whose first hidden state is the sum of the usual input projection $W_{\text{in}}x_t$, a dedicated self-conditioning projection $W_{\text{sc}}\tilde{x}_0$, and the position and timestep embeddings. Gradients only flow through the second pass. On the other half of training steps, $\tilde{x}_0$ is set to zero, which matches the test-time first denoising step. During sampling, the estimate $\hat{x}_0$ from each DDIM step is detached and used as $\tilde{x}_0$ at the next step.

3.4 Attribute Classifier and Guided Sampling

We jointly train a lightweight attribute classifier $g_\phi: \mathbb{R}^d \rightarrow \mathbb{R}^K$ that operates on a target-masked pooled average of the noisy embeddings,

$$z = \frac{\sum_i m_i x_{t,i}}{\sum_i m_i}, \quad g_\phi(z) = W_2 \text{SiLU}(W_1 z), \quad (3)$$

where m is the binary target mask. The classifier is trained with cross-entropy against the discrete value of a chosen slot. For `food`, the label space is constructed once over the training split: index 0 is reserved for `<missing>` (the MR does not contain the slot), and the remaining $K - 1$ indices enumerate the sorted set of observed `food` values. The classifier loss is added to the diffusion loss with weight $\lambda=0.1$:

$$\mathcal{L} = \mathcal{L}_{\text{diff}} + \lambda \cdot \text{CE}(g_\phi(z), y). \quad (4)$$

Crucially, g_ϕ is trained on *noisy* pooled embeddings, exactly the input it will see at sampling time, so the classifier learns to score partially denoised states rather than only clean text.

At sampling, before each DDIM step we take a gradient ascent step on the target log-probability of the desired class y^* :

$$x_t \leftarrow x_t + s \cdot \frac{\nabla_{x_t} \log p_\phi(y^* | x_t)}{\|\nabla_{x_t} \log p_\phi(y^* | x_t)\|_2}, \quad (5)$$

with guidance scale $s=0.5$. The gradient is zeroed out on the prefix positions, and the prefix is re-clamped after the update, so the MR remains untouched.

3.5 EMA Weights

We maintain an exponential moving average of the denoiser parameters with decay 0.999 throughout training and use the EMA copy for both validation and checkpoint export. EMA is standard practice in diffusion training [8,14]; we found it materially reduced sample-quality variance across consecutive epochs at our model scale.

3.6 Baselines

GPT-2 (small) [17]: causal LM fine-tuned on “MR <|sep|> reference” with the EOS token used as both pad and end. At decoding we use nucleus sampling with $p = 0.95$ and temperature 1.0, which trades a small BLEU drop for higher distinct- n — the diversity-oriented setting that gives diffusion the strongest possible baseline to match on lexical variety. **T5-small** [18]: seq2seq with source=MR, target=reference; beam search with $b=4$. T5 with beam search is the strongest pure fluency baseline.

4 Experimental Setup

Dataset. E2E NLG cleaned [15,5]; 33,525 / 4,299 / 4,693 train/val/test pairs. MR slots: name, eatType, food, priceRange, customerRating, area, familyFriendly, near. MRs are flattened to a “key: value | key: value” string before tokenization with the **bert-base-uncased** WordPiece tokenizer.

Training. All models are trained on a single A100 GPU (Google Colab Pro+). **Diffusion-LM**: 20 epochs, batch size 64, AdamW, learning rate 1×10^{-4} , linear warmup over 1000 steps, gradient clipping at norm 1.0, $T=2000$ diffusion steps. **Baselines**: 5 epochs, batch size 32; learning rates 5×10^{-5} (GPT-2) and 3×10^{-4} (T5). Maximum sequence length is 96 tokens with prefix length 32.

Decoding. GPT-2: nucleus sampling ($p = 0.95$, $\tau = 1.0$). T5: beam search with $b = 4$ and length penalty 1.0. Diffusion-LM: DDIM [19] with 200 steps and a rounding interval of 10 steps, self-conditioning enabled, classifier guidance scale 0.5 on the **food** slot. Generation is batched (16 examples per batch) to keep inference cost tractable.

Metrics. *Fluency*: BLEU [16] via **sacrebleu**, ROUGE-L [12] (F1). *Diversity*: distinct-1 and distinct-2 [9], the ratio of unique n -grams to total n -grams in the generated corpus. *Controllability*: slot-fill accuracy, the fraction of MR slot values that appear verbatim (case-insensitive) in the generated output. Slot-fill accuracy is a permissive lexical-match metric — it does not penalize paraphrase but does catch slot omissions, which is the failure mode E2E benchmarks have been historically designed to surface.

Table 1. Main results on E2E NLG test set (4,693 examples). BLEU and ROUGE-L are reported on a 0–100 scale; distinct- n on a 0–1 scale.

Model	BLEU	ROUGE-L	distinct-1	distinct-2	Slot	Acc.
GPT-2 (nucleus)	...	...	...	...	...	...
T5-small (beam-4)	...	...	...	...	...	...
Diffusion-LM	...	...	...	...	...	...

Table 2. Ablations on Diffusion-LM. The sqrt schedule with prefix length 32 and 200 DDIM steps is the default configuration reported in Table 1. Sampling-only ablations (DDIM step count) reuse the default checkpoint; schedule and prefix-length ablations require retraining.

Variant	BLEU	ROUGE-L	distinct-2	Slot	Acc.
sqrt schedule (default)	...	...	...	...	...
cosine schedule	...	...	...	...	...
DDIM 50 steps	...	...	...	...	...
DDIM 100 steps	...	...	...	...	...
DDIM 200 steps	...	...	...	...	...
prefix length 16	...	...	...	...	...
prefix length 32	...	...	...	...	...
prefix length 48	...	...	...	...	...

Reproducibility. All configurations are stored as YAML files under `configs/`. A 10-second CPU smoke test (`tests/smoke_test.py`) exercises the MR parser, all three noise schedules, the Diffusion-LM forward/backward/sample paths with prefix clamping, self-conditioning, classifier guidance, and the full metric pipeline. The training script writes both raw and EMA checkpoints per epoch and selects the best-val-loss checkpoint for evaluation.

5 Results

Table 1 reports the test-set comparison between the three systems on all five metrics; Table 2 reports the ablations on Diffusion-LM. We organize the discussion around three questions: *(i)* how do the three paradigms (AR with beam search, AR with nucleus sampling, diffusion) trade off fluency, diversity, and controllability? *(ii)* how sensitive is Diffusion-LM to its main hyperparameters — noise schedule, DDIM step count, and prefix length? and *(iii)* which of the four design choices in our base model (prefix clamping, target masking, BERT-init, EMA) and the two add-ons (self-conditioning, classifier guidance) leave the largest fingerprint on the metrics? The Analysis (Section 6) and Discussion (Section 7) sections answer these in turn.

6 Analysis

Expected slot-level failure modes. The **food** and **near** slots have the largest open vocabularies in E2E and are known to be the hardest to recover lexically [5]: they are most often paraphrased (“Indian” → “curry house”) or omitted entirely. We therefore expect T5 with beam search to be the most slot-faithful overall but the least lexically varied, GPT-2 with nucleus sampling to be the most varied but prone to hedging on contradictory MRs (e.g. **priceRange**[high] with **customerRating**[low]), and Diffusion-LM with classifier guidance to raise the **food**-slot accuracy specifically while leaving the other slots controlled by prefix clamping alone. These predictions are verified against the per-example outputs and broken down by slot in the companion notebook `notebooks/error_analysis.ipynb`.

Rounding interval and step-count interaction. The rounding step in Diffusion-LM is an early-commitment to discrete tokens: once the embedding is projected through the embedding matrix and re-embedded, it locks the local context for subsequent denoising steps. With very few DDIM steps the rounding fires before the surrounding target is fully denoised, which can produce locally ungrammatical fragments. We treat the rounding interval as a hyperparameter and report sensitivity to it in the ablation results.

Classifier guidance scale. The guidance scale s trades off attribute fidelity against on-manifold fluency: at large s the gradient update pulls the denoiser away from the training distribution, hurting BLEU; at very small s the update is indistinguishable from no guidance. We selected $s=0.5$ via a coarse validation grid search before reporting test-set numbers; we do not tune s further on the test split.

7 Discussion

Hard vs. soft conditioning. Prefix clamping is hard conditioning: the MR is never noised and never moved. Classifier guidance is soft: the denoiser is pushed toward the target class but the prefix remains the ground truth. The two are complementary. Hard clamping is the right choice for slots that must appear verbatim (e.g. restaurant **name**); soft guidance is the right choice for slots that benefit from paraphrase (e.g. **food**, which can be realized as “Italian”, “Italian cuisine”, “Italian dishes”). Our implementation supports both simultaneously.

Diversity-fluency trade-off. The three systems occupy distinct points on the diversity-fluency frontier. T5 beam search optimizes for likelihood and collapses to templates, GPT-2 nucleus sampling optimizes for diversity at a small cost in BLEU, and Diffusion-LM occupies an intermediate point: the iterative refinement produces more varied surface forms than a beam search, but the rounding step pulls back toward high-probability tokens at every iteration.

Effect of self-conditioning. Self-conditioning gives the denoiser access to its previous x_0 prediction. In small models this is particularly useful because each individual denoising step has limited capacity; sharing information across steps amortizes that limit. The cost is modest: a single linear projection from the embedding dimension to the denoiser hidden dimension, plus one extra forward pass during training half the time.

Embedding initialization matters. Random initialization of the token embedding table leaves the rounding cross-entropy to bootstrap a meaningful geometry from scratch, which at small scale tends to collapse the embedding space into a few clusters. Initializing from a pretrained BERT embedding table gives the rounding objective a near-orthogonal starting geometry and was, in our experience, the single change with the largest effect on output quality at the model and dataset scale we operate at.

Limitations. Three constraints shape what our results can show. (i) *Model scale.* Our denoiser has six layers and roughly 50M parameters, an order of magnitude smaller than the original Diffusion-LM. At this scale, performance is bounded by capacity rather than algorithm. (ii) *Benchmark choice.* E2E NLG is heavily templated: a small, mostly-closed slot vocabulary, short references (~ 20 – 30 tokens), and naturally low human-reference lexical diversity (distinct-2 ~ 0.05 – 0.10). This structurally favors AR models with beam search, which can essentially memorize the dominant templates. A benchmark with multiple valid outputs per input and more open vocabulary (e.g. CommonGen [11] or ROCStories [13]) would make the iterative-refinement and plug-and-play-guidance advantages of diffusion more visible in the metrics; we discuss this further as future work. (iii) *Metric.* Slot-fill accuracy is a permissive lexical-match metric — a model that paraphrases “Italian” as “Italian cuisine” is penalized only when the two strings happen to disagree. A more faithful controllability metric would parse the generated output back into an MR and compare slot-by-slot.

Ethical considerations. Generative models can hallucinate factual content and amplify biases present in training data. On E2E the risk is bounded (synthetic restaurant descriptions in a controlled vocabulary), but the framework is general: deployed at scale on open-domain text it would require bias auditing and content filtering. We do not release a trained checkpoint with this work.

8 Conclusion

We presented an implementation and empirical study of prefix-conditioned Diffusion-LM for controllable text generation on E2E NLG, augmented with self-conditioning, EMA weights, BERT-initialized embeddings, and a jointly trained attribute classifier used for gradient-based guidance during DDIM sampling. The model combines hard prefix conditioning, which enforces attribute control by construction, with soft classifier guidance, which trades fluency for finer per-slot control. We

compared against fine-tuned GPT-2 and T5 baselines along fluency, diversity, and controllability axes, and ablated noise schedule, DDIM step count, and prefix length.

Three concrete next steps follow directly from our limitations. First, the single-slot classifier should be extended to a multi-head variant that guides all eight E2E slots simultaneously, allowing per-slot guidance scales. Second, the lexical slot-fill metric should be replaced by a parser-based MR-recovery score that credits paraphrase. Third, the framework should be evaluated on a benchmark whose structure rewards iterative refinement and diversity — CommonGen [11], with multiple valid outputs per concept set, or ROCStories [13], with longer creative continuations — where the diffusion paradigm has more room to differentiate from beam search.

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